

Optical absorption spectra of gold nano-clusters in potassium borosilicate glass

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Abstract

Optical absorption spectra for gold particles grown in two different potassium borosilicate glasses at 600 °C were measured. In the more viscous glass, only extremely small particles formed by nucleation of gold atoms in the glass. In the less viscous glass, the particles nucleated and grew to larger sizes, forming a plasmon absorption band. Particles sizes were calculated from the optical absorption spectra, and the diffusion coefficient of gold in the glass was also calculated. A transmission electron micrograph (TEM) showed a distribution of particle sizes resulting from both nucleation and growth of the particles. The influence of this size distribution on kinetics of growth and the optical absorption spectra was discussed.

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1. Introduction

Gold nano-clusters have been studied a long time for their optical properties in the UV/visible light range. Band-to-band transitions in the shorter wavelengths and a plasmon oscillation centered on 500–520 nm absorb incoming light in the blue-green region of the visible spectra [1], which is why nano-clusters of gold appear red [1–4].

Gold nano-clusters grown in glass have been of particular interest for their applications in optical devices such as optical filters. Glass is a good material to hold the gold particles, because it is transparent in the visible light range and provides a protective matrix for the particles.

A challenge with growing the clusters has been controlling the size distribution. Growing small clusters in a controlled way is rather difficult. The resulting absorption spectra are not completely understood, as size dis-

tributions and the band-to-band region that overlaps the plasmon region make it hard to interpret the spectra.

Controlling growth is also difficult for another reason. Particles, even if successfully grown, are difficult to characterize because they are hard to resolve in transmission electron microscopy (TEM) and scanning electron microscopy (SEM). The smallest clusters are difficult to see in SEM (at 1–20 nm in diameter), and in TEM particles less than 5 nm in diameter are obscured by glass irregularities. TEM sample preparation is difficult, and not an exact science where glass is concerned but with time and practice it is possible [5].

In the present work gold particles below 10 nm in diameter were grown in two different silicate glass compositions with different rates of particle growth. The optical spectra of particles of different sizes were measured from 0.3 to 0.9 μm wave length. The effect of particle size on the plasmon absorption band was followed by separating out the free electron contribution to this band, showing how particle size modified the optical properties of the gold from their bulk values. Different nucleation and growth kinetics for the two glass compositions were explored.

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2. Experimental methods

The samples for this research were created from powders of the different constituents and a gold chloride (hydro-auric tetra chloride, $\text{HAuCl}_4 \cdot x\text{H}_2\text{O}$). The two different glass compositions are shown in Table 1. The rate of growth of the gold particles in glass B was faster than in glass A, because glass A had more silica and less K_2O than glass B, which led to lower viscosity for glass B.

The gold chloride was hand mixed with ethanol, which was then mixed with the oxide constituents. This slurry was then dried and the mixture was ground with a mortar and pestle to mix it further. About 50 g of mixture was then measured into a Coors high-purity alumina crucible of 100 ml volume. Different powder mixtures were subjected to different preheat-treatments, which are listed in Table 2.

The samples were then melted in the same CM Inc. Rapid Temp Furnace as the preheat-treatments, so samples remained in the furnace. Melting was done at $\sim 1500^\circ\text{C}$ for 2 h. Cooled samples were transparent only if they had been heated for at least 2 h. The molten samples were poured onto a graphite slab to quench them quickly, so there are no uncontrolled particle growth. Controlled growth is possible if one starts with a transparent glass with no clusters, forms small nuclei by irradiating the samples with UV light or heating them, and then grows a controlled number of clusters through heat-treatment [3,6,7].

After quenching, the transparent samples were subjected to an additional heat-treatment followed by a furnace cool to anneal out any internal stresses created from the rapid quenching. The much more viscous A samples were heated for 1 h at 600°C while the B samples were heated for 10 min at 500°C . The absence of internal stresses was verified with a polariscope.

The samples were prepared by polishing on both sides to a uniform thickness of $\sim 3\text{ mm}$ and irradiated with UV light for 2 min to form particles. B6 was the only sample not irradiated.

Table 1

Weight percentages of different constituents used in making batches of B and A samples

	SiO_2	B_2O_3	K_2O	Al_2O_3	Au
B	62.2%	9.37%	25.8%	2.58%	0.019%
A	73.8%	9.37%	14.2%	2.60%	0.0195%

Table 2

Preheating conditions for powders

Sample	A1 B1	A2 B2	A3 B3	B4	A4 B5	A5 B6
Time	0	0	1 hour	1 hour	4 hours	4 hours
Temperature	—	—	850°C	850°C	850°C	850°C

After irradiation and heating, absorption spectra were measured. An initial spectrum from the colorless glass was subtracted from the subsequent spectra from samples containing gold particles. The range chosen for optical absorption spectra was 300 nm to 900 nm. Below 300 nm the glass absorption obscured absorption from the particles. The scan rate was 100 nm/min, and data points were recorded every 5 nm. Samples were heated for different times (which were each followed by furnace cools) in a H_2 atmosphere at 600°C . H_2 was used as an atmosphere to reduce the gold to atoms; in O_2 the gold might remain ionized.

3. Experimental results

B samples after heating 5 h turned yellow. Except for the first batch of B3 and 4 (4 h preheat of batch at 850°C), all samples turned red within 30 h of heat-treatment. The samples preheated for 1 h (B5 and 6) turned red very quickly (5–15 h), and became a deep and brilliant red after longer annealing times. The more viscous A samples were initially yellow, and became a darker yellow and then brownish at longer annealing times.

Spectra for two of the B and A samples are shown in Figs. 1 and 2, respectively. The spectrum of the initial clear glass was subtracted from subsequent spectra, so the spectra shown are only due to the gold. One can see that the yellow color of the A samples and short-time B samples results because of absorption that is roughly linear with wavelength below about 600 nm and increasing to 300 nm.

For annealing times longer than about 5 h, the absorption spectra for A glass do not change shape, as shown in Fig. 3, in which spectra for 5 h and 90 h anneal-

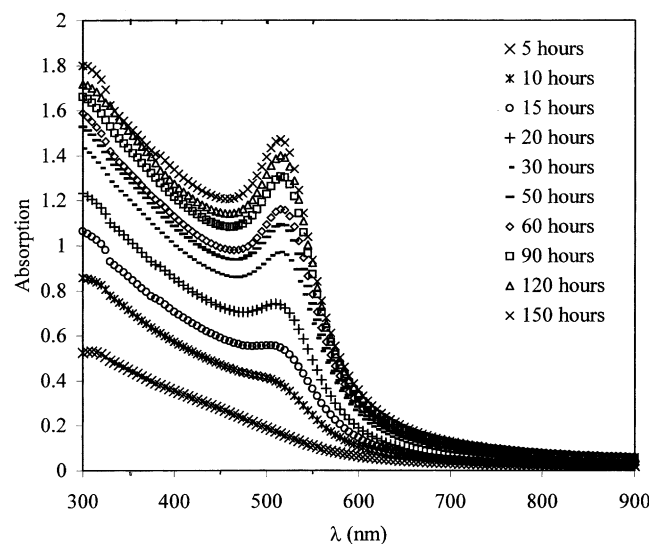


Fig. 1. Optical absorption spectra of glass B5 containing small gold particles heated for different times at 600°C .

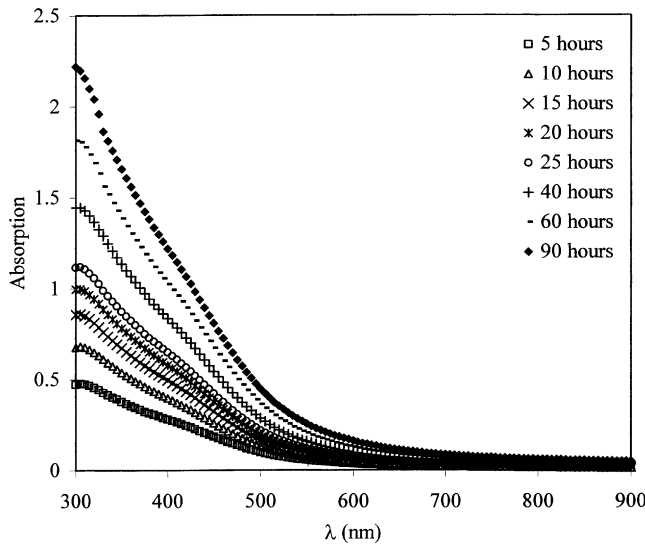


Fig. 2. Optical absorption spectra of glass A5 containing small gold particles heated for different times at 600°C.

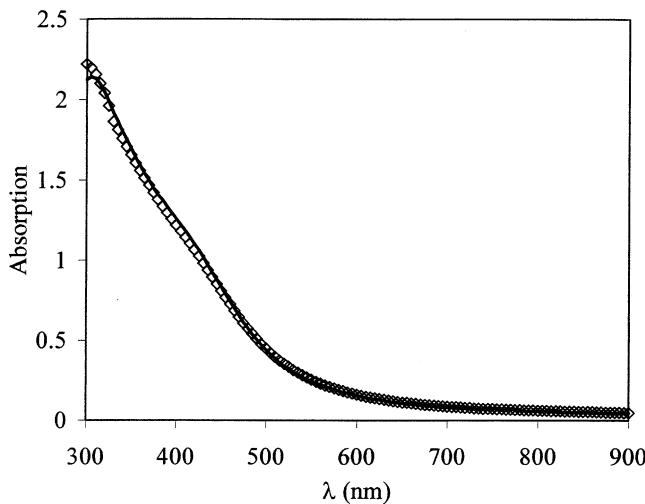


Fig. 3. Comparison of optical absorption spectra from Fig. 2. $\diamond$ 90 h spectrum, line, 5 h spectrum multiplied by 4.5.

ing times are compared and differ only by a multiplying factor that is constant with wavelength.

A TEM picture of sample B6 after 120 h at 600°C in Fig. 4. The particles range in radius from about 3 to 8 nm.

4. Optical absorption of gold spheres

The optical properties of a solid are described here in terms of the complex dielectric constant $\epsilon^* = \epsilon_1 - i\epsilon_2$. The optical properties of noble metals can be separated into two parts, one related to free electrons and the other to band-to-band transitions of electrons localized on the atoms [1]. In gold there is a transition in ϵ_1 values

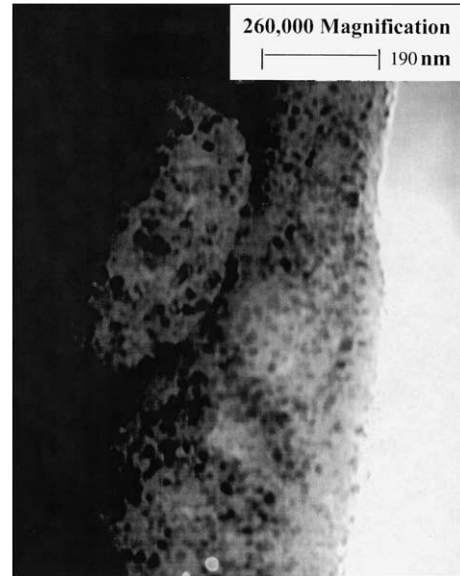


Fig. 4. TEM of sample B6 after heating 120h at 600°C. Crushed sample, Philips CM12 electron microscope.

at a wavelength of about 0.48 μm ; at higher wavelengths, ϵ_1 is proportional to λ^2 as expected for free electrons [1,8,9]. The transition in ϵ_2 occurs at a longer wavelength of about 0.65 μm , above which ϵ_2 is proportional to λ^3 . From the measurements of Theye on thin gold films [9].

$$\epsilon_2 = 0.1 + 3.4\lambda^3, \quad (1)$$

with wavelength λ in μm .

As the gold particles become smaller, the particle size becomes similar to the mean-free-path of the free electrons in the gold, and the plasmon absorption band becomes broader. The values of ϵ_1 are unaffected by particle size, and the values of ϵ_2 can be corrected for the reduction of the bulk mean free path l_o of the electrons [10].

$$\frac{1}{l_{1e}} = \frac{1}{R} + \frac{1}{l_o}, \quad (2)$$

in which R is the radius of a spherical particle and l_e is the effective mean free path of the electrons in the small sphere. Combining Eqs. (1) and (2)

$$\epsilon_2 = 0.1 + 3.4(l_o/R + 1)\lambda^3. \quad (3)$$

A value of 20 nm for l_o is used from the measurements of Theye [9]; this value is smaller than the mean free path of 40 nm for electrons in bulk gold deduced from electrical measurements [1], probably because of additional electron scattering from the grain boundaries in the gold films of Theye.

Mie first quantitatively explained the mathematics of a plasmon resonance [2]. Optical absorption characteristics of red (gold filled) glass demonstrate contributions from band-to-band transitions, as well as a collective plasmon resonance. Band-to-band transitions are

characteristic of localized electrons, and the plasmon resonance is due to the collective resonance of the free electrons in small particles. The width and height of the absorption peak is dependent on the size, shape and quantity of particles. The dependency on size can be explained by the Drude free electron theory [1]. Particles smaller than the mean free path of the electrons scatter the electrons from the particle surface, giving a broader plasmon peak, whereas large particles give a sharper and higher peak [6,11]. The shape of the peaks is usually a Lorentzian curve [10]. Much research on plasmon resonances is reviewed in Ref. [11].

Mie's mathematical solution for absorption and scattering from spherical metal clusters has been proven by many experiments. His equations are long and their derivations are longer; only the equation pertinent to the present work is given here. The equation for a collection of spherical metal clusters that are much smaller than the wavelength of light in a matrix is

$$\alpha = 18\pi r^3 NV \frac{1}{\lambda} \frac{\epsilon_2}{(\epsilon_1 + 2n^2)^2 + \epsilon_2^2}, \quad (4)$$

in which α is the effective absorption coefficient of the composite, n is the refractive index of the matrix (glass, 1.5 in this work), V is the volume of a single particle, N is the number of particles per unit volume, and λ is the wave length of light. This equation assumes all particles are the same size.

5. Discussion

There is a striking difference between the optical absorption spectra in Figs. 1 and 2 for the two different glasses A and B, which do not differ much in composition. The spectra for sample B in Fig. 1 show the development of the plasmon peak at about 0.52 μm wavelength as the heating time increases, corresponding to an increase in particle size [2,6,11,12]. The 5 h spectrum is about the same for the two glasses, and the shape of this spectrum remains the same for sample A as it is heated at 600 °C (Figs. 2 and 3), whereas for sample B the shape of the spectrum changes with development of the plasmon band. A possible explanation for these differences is given in the next paragraphs.

The 5 h spectrum in Figs. 1 and 2 is characteristic of very small gold particles, less than about 0.5 nm in radius, containing only about 30 gold atoms [6,12]. In this spectrum the plasmon peak has spread so wide that it no longer shows a maximum.

When the glass is heated at melting temperatures the gold goes into solution in the molten glass as ions. When the glass is quenched rapidly enough the ions remain in the oxidized state. As the glass is heated in hydrogen at 600 °C the gold ions are reduced to gold atoms [7]. The first step in formation of particles is nucleation of gold

by fluctuations of the concentration of gold atoms. This process is apparently faster than growth, perhaps because of non-uniformities in our glasses (randomness) of gold concentration. This nucleation is homogeneous, rather than heterogeneous, as shown by the small particle size and consequent high number density of gold particles. The ultraviolet irradiation did not induce nucleation; the spectra of sample B6, which was not irradiated, grew with the same time dependence as for the irradiated samples. Ultraviolet radiation does nucleate gold particles when the glass contains a nucleating agent such as cerium oxide [3].

Thus the 5 h spectrum results from the nucleation of gold particles, and this process has about the same rate in the two glass compositions, because it does not involve long-range diffusion of gold atoms. In glass A nucleation continues as the glass is heated at 600 °C, as shown by the spectra that have the same shape during heating (Figs. 2 and 3). The long-range diffusion in this glass is too slow to grow the nucleated particles. In glass B the nucleated particles grow by long-range diffusion of the gold atoms through the glass.

An approximate diffusion coefficient for the gold atoms can be calculated from the kinetics of the growth of the particles. The diffusion-controlled growth of a collection of spherical particles of the same size follows the empirical equation [13].

$$W = 1 - \exp(kt^{3/2}), \quad (5)$$

in which W is the volume fraction of total growth.

$$W = \frac{NVC_p}{C_i}, \quad (6)$$

N is the number of particles per unit volume, C_p is the density of gold (19.3 g/cm³) and C_i is the concentration of gold atoms in the glass, which is 4.75(10)⁻⁴ g/cm³ if all the gold is reduced. Eq. (5) is close to the rigorous equation up to W about 0.8 [13]. The experimental W values calculated from the optical absorption in Fig. 1 at 0.45 μm wave length fit this equation to about $W = 0.75$, showing that the growth of the gold particles in this glass is diffusion-controlled. The parameter k is given by

$$k = \frac{4\pi N}{3} \left(\frac{C_i}{C_p} \right)^{1/2} (2D)^{3/2}, \quad (7)$$

in which D is the diffusion coefficient. In the early stages of particle growth before the diffusion fields of the particles overlap the radius r of a particle is given by [13]

$$r^2 = 2C_i D t / C_p. \quad (8)$$

If the volume fraction W and particle radius r after diffusion time t are known, the diffusion coefficient D and particle number density N can be calculated from Eq. (5)–(8). From the spectrum for glass B5 after 20 h at 600 °C (Figs. 1 and 7) the calculated particle radius

(see below) is 3.8 nm and $W = 0.61$. Then from Eqs. (5)–(8), $D = 6.2(10)^{-14} \text{ cm}^2/\text{s}$ and $N = 6.5(10)^{13} \text{ particles/cm}^3$. This diffusion coefficient of gold atoms is smaller than that for gold in a sodium aluminum silicate glass of $4.3(10)^{-12} \text{ cm}^2/\text{s}$ (7), probably because the viscosity of the latter glass is lower than the viscosity of glass B.

Eqs. (5)–(8) are valid for particle growth when all particles are the same size. Since there is probably some continuing nucleation in glass B5 as growth proceeds, there is a distribution of particle sizes in the samples. The TEM in Fig. 4 shows this distribution for sample B6. Thus the above calculations are approximate only, and refer to some average size of the growing gold particles.

The free electron contribution to the plasmon peak at 522 nm wavelength can be identified by subtracting the 5 h spectrum from the spectra after some particle growth, as shown in Fig. 5. The intensity of the 5 h spectrum was equated to the spectrum of the larger particles at 300 nm to make this subtraction. The results of this subtraction for the spectra in Fig. 1 are shown in Fig. 6. The resulting plasmon bands are nearly symmetrical, and fit closely with the spectrum calculated from the free-electron contribution to the optical properties ϵ_1 and ϵ_2 . The values of ϵ_2 were taken from Eq. (3), as modified for a small particle size R . The values of ϵ_1 are given by an extrapolation of the free-electron contribution at wavelengths longer than $0.48 \mu\text{m}$ [8,9]

$$\epsilon_1 = 10 - 58\lambda^2, \quad (9)$$

with the wavelength λ in μm . The calculated spectrum for sample B heated 20 h at 600°C is compared with the experimental one in Fig. 7; the fit is good. The radius used for this fit was 3.8 nm. Another example is in Fig. 8 for sample B6 after 120 h at 600°C , the same sample as

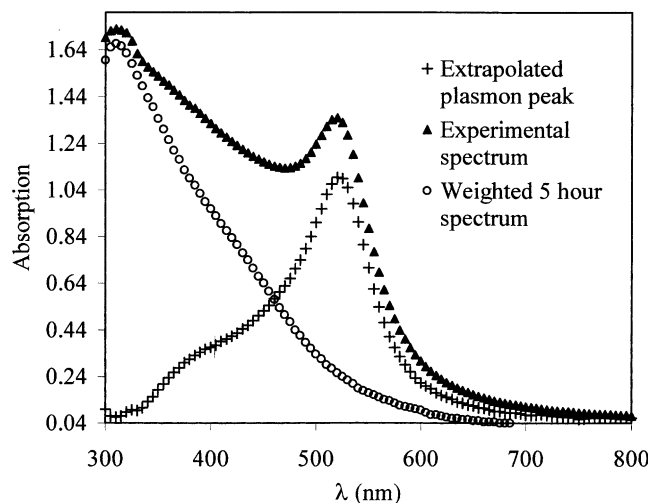


Fig. 5. Subtraction of 5 h spectrum from 120 h spectrum for sample B6. The 5 h spectrum was weighted (multiplied) to nearly match the 120 h spectrum at 300 nm.

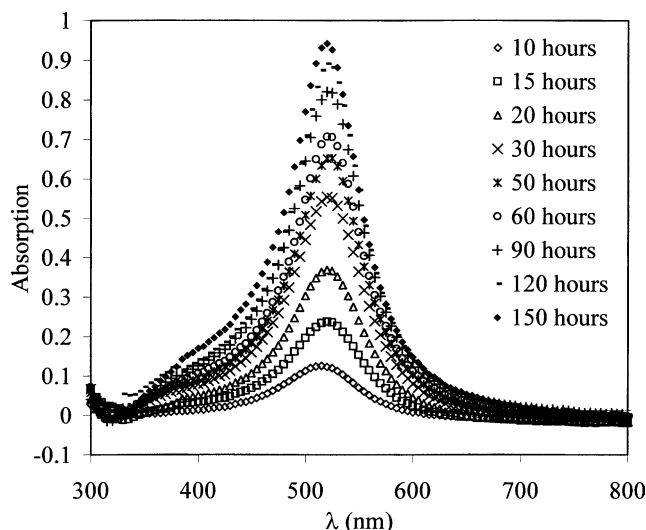


Fig. 6. Optical absorption spectra of sample B6 after subtraction of the 5 h spectrum (see Fig. 4 for an example) for different heating times at 600°C .

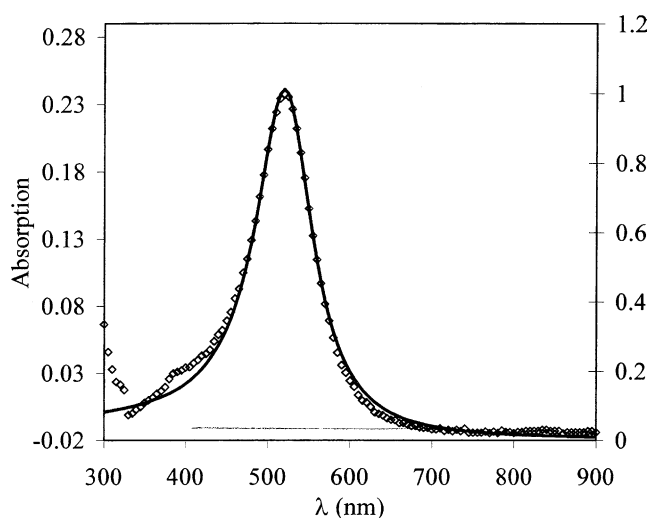


Fig. 7. Points, optical absorption spectrum of sample B5* from Fig. 1 after 20 h at 600°C after subtraction of 5 h spectrum multiplied by two. Line, spectrum calculated from Eqs. (3) and (4) with a particle radius of 3.8 nm.

shown in Fig. 4. The calculated absorption spectrum in Fig. 8 was for a particle radius of about 5.3 nm. The particle radii in Fig. 4 range from about 3 to 8 nm, so this size calculated from the optical results is reasonable, and refers to some average radius.

The distribution in the particle sizes because of both growth and continuing nucleation in the B glass sample mean that the calculations of the last few paragraphs are approximations, because the kinetic and optical equations are derived for particles all of the same size. How to average the particle size distributions to correspond to the calculated particle sizes from the optical equations and the diffusion coefficients is complicated,

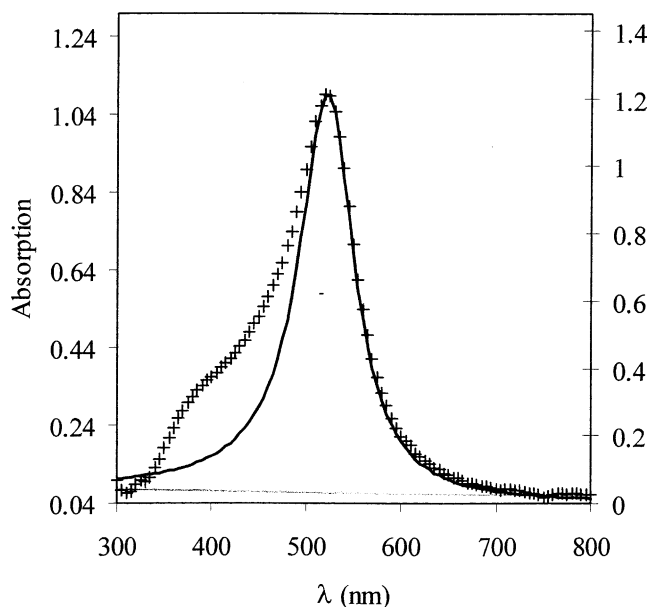


Fig. 8. Points, optical absorption spectrum of sample B6 from Fig. 5 after heating 120 h at 600°C, after subtraction of the weighted 5 h spectrum. Line, spectrum calculated from Eqs. (3) and (4) with a particle radius of 5.3 nm.

and requires a detailed knowledge of the size and spatial distributions of the particles. An example of a simplified calculation of the absorption spectrum is described in the next paragraph.

To examine the influence of the size distribution, a square of side 10 cm was chosen at random from the micrograph of Fig. 4, and spectra were calculated for each individual particle. The sum of all these spectra weighted for the particle volume were normalized, and the fit to the experimental spectrum is shown in Fig. 9. The fit is only slightly better than that in Fig. 8 for the assumption of one particle size, showing that that assumption can give useful comparisons.

In many previous studies of growth of noble metal particles and their optical absorption there have been considerable distributions in the particle sizes [11]. The present work shows that if this distribution is not too broad, meaningful deductions about particle size and the mechanisms of particle growth are possible.

6. Conclusions

A small change of composition between two potassium borosilicate glasses produces a large change in the kinetics of formation and optical properties of gold particles grown in the glasses. In the more viscous glass, nucleated particles do not grow much, because the diffusion coefficient of gold atoms in the glass is too small. In

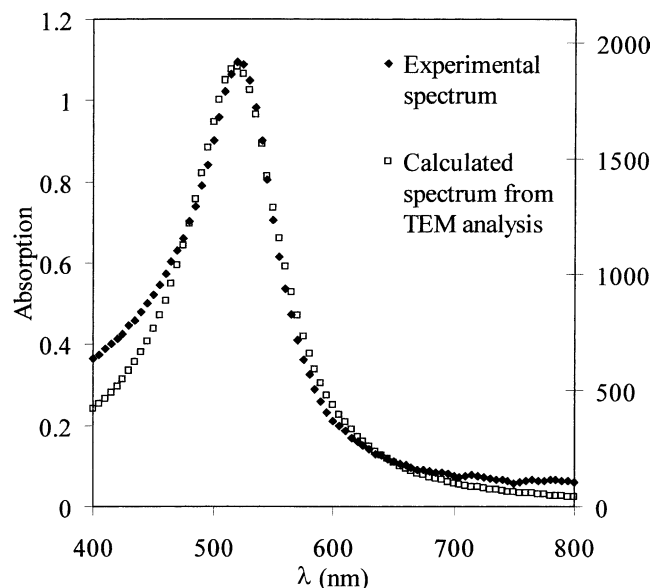


Fig. 9. (♦) Points, optical absorption spectrum of sample B6 after heating at 600°C for 120 h after subtraction of weighted 5 h spectrum, from Figs. 5 and 8. This is the same sample as shown in Fig. 4. (□) Points, spectrum calculated from Eqs. (3) and (9) as described in the text, using the distribution of particle sizes from Fig. 4.

the less viscous glass the nucleated particles grow and a plasmon absorption band develops. Although there is a distribution of particle sizes in the less viscous glass, an average particle size can be calculated from the optical absorption spectra, and this size is consistent with particle sizes seen in a TEM. A reasonable diffusion coefficient of gold in the glass was also calculated from the particle sizes and absorption spectra.

References

- [1] N.F. Mott, H. Jones, *The Theory of the Properties of Metals and Alloys*, Dover, 1958, p. 105ff.
- [2] G. Mie, *Ann. Physik.* 25 (1908) 337.
- [3] S.D. Stookey, *J. Am. Cer. Soc.* 43 (1949) 246.
- [4] H.C. van de Hulst, *Light Scattering by Particles*, John Wiley, NY, 1957, p. 270.
- [5] R.H. Doremus, A.M. Turkalo, *J. Mat. Sci.* 11 (1976) 903.
- [6] R.H. Doremus, *J. Chem. Phys.* 40 (1964) 2389.
- [7] R.H. Doremus, in: M.K. Reser, G. Smith, H. Insley (Eds.), *Symposium on Nucleation and Crystallization in Glasses and Melts*, The American Ceramic Society, 1962, p. 119.
- [8] W. Otter, *Z. fur. Physik.* 161 (1961) 163.
- [9] M.L. Theye, *Phys. Rev. B* 2 (1970) 3060.
- [10] W.J. Doyle, *Phys. Rev.* 111 (1958) 1067.
- [11] U. Kreibitz, M. Vollmer, *Optical Properties of Metal Clusters*, Springer, Berlin, 1995.
- [12] R.H. Doremus, P. Rao, *J. Mater. Res.* 11 (1996) 2834.
- [13] R.H. Doremus, *Rates of Phase Transformations*, Academic, San Diego, 1985, p. 22.